



**IUBAT - International University of
Business Agriculture and Technology**

Project Proposal

Course Code: CSC 440

Submitted to:

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Submitted by:

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Section: G

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1. Project Name

E-Shop Manager — A Mini E-commerce & Inventory Management System

2. Group Members' Names and IDs

SL	Name	Student ID
1	M Tafsirul Islam Chowdhury	23303354
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3. Expected Features

SL	Feature	Description
1	Role-Based Login System	Separate Admin and Customer login using JWT-based authentication; role enum (Admin, Customer) controls access to features.
2	Product & Inventory Management	Admin can add, update, and delete products (name, price, stock, category). Low-stock alerts when quantity falls below threshold.
3	Cart & Order Processing	Customers browse products, add/remove items in cart, and place orders. Order and cart items are calculated and stored per session.
4	Discount & Price Calculation Engine	Category-wise and order-type-wise (Regular / Premium / Bulk) discount rules applied through OOP polymorphism (method overriding).
5	Order History & Invoice	Customers can view past orders; a simple invoice summary is generated per completed order.
6	Search & Filter	Products can be searched by name and filtered by category on the frontend.

4. How to Implement (Tools & Technologies)

Layer	Tools / Technologies
Frontend	Next.js (React), Tailwind CSS — product listing, cart UI, checkout, and order-history pages; consumes backend REST API via fetch/axios.
Backend	ASP.NET Core Web API (C#) — RESTful API with Controllers and Services; OOP models (Product, Order, Cart, Customer) demonstrating inheritance, polymorphism, and encapsulation (e.g., RegularOrder / PremiumOrder / BulkOrder overriding ApplyDiscount()).
Database	MongoDB, accessed via the official MongoDB.Driver package in ASP.NET Core. Collections: Products, Orders, Customers, Carts.
Authentication	JWT (JSON Web Token) based authentication with role claims (Admin / Customer).
Version Control	Git & GitHub for source control and team collaboration.

Architecture: Next.js frontend communicates with the ASP.NET Core Web API backend over REST (JSON); the backend persists data in MongoDB.